

Predicting Frequent Depression Among U.S. Adults Using Pooled NHIS 2022-2024 Data: An Interpretable Survey-Weighted Risk-Stratification

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Project type: Prediction, association, and public-health risk stratification; not causal inference

Abstract

Frequent depression is a clinically meaningful and policy-relevant indicator in population health surveillance. This project developed an interpretable prediction workflow for frequent depression among U.S. adults using pooled 2022-2024 National Health Interview Survey (NHIS) Sample Adult data. Frequent depression was defined as reporting depressed mood daily or weekly, versus all lower-frequency responses. The analytic sample included 87,809 adults with valid depression-frequency responses, and survey weights were pooled as WTFA_A / 3. Candidate predictors were organized into conceptual domains: health burden, socioeconomic hardship and access barriers, demographic/social context, and timing. A broad candidate set was narrowed to a 12-variable presentation model, then three 12-variable models were compared: weighted logistic regression, LASSO logistic regression, and random forest. The final model was the 12-variable weighted logistic model because it had nearly identical discrimination to LASSO, lower calibration error than random forest, and the clearest interpretation for public-health use. In held-out test-set evaluation, the 12-variable logistic model achieved weighted AUC approximately 0.787, Brier score approximately 0.082, calibration slope approximately 0.94, and expected calibration error approximately 0.011. Calibration was examined using fixed predicted-risk groups, separating the population distribution from the comparison of mean predicted risk and observed prevalence. Threshold analysis showed that the highest-risk 10%, 20%, and 30% of the weighted adult population captured approximately 35.8%, 56.0%, and 68.2% of weighted frequent-depression cases, respectively. A secondary prescription-user analysis found that overlapping medication-cost barriers and mental-health care cost barriers marked a high-risk subgroup. These results support use of the model for outreach prioritization, screening invitations, navigation support, and resource planning, while avoiding causal or diagnostic claims.

1. Introduction

Depression-related symptoms are common in population health data and are closely connected to social hardship, access barriers, and overall health burden. For public-health practice, the goal is often not to diagnose depression from survey variables, but to identify groups who may benefit from screening, outreach, referral, or navigation support. National survey data are well suited for this type of risk-stratification work because they include health status, socioeconomic context, health-care access, and medication affordability measures in a large, representative sample.

This project used the 2022-2024 NHIS Sample Adult files to predict frequent depression among U.S. adults. The outcome was defined from DEPFREQ_A as daily or weekly depressed mood versus all lower-frequency responses. The primary goal was to create an interpretable, survey-weighted prediction workflow that could be explained in public-health language. The secondary goal was to examine whether prescription users with multiple medication-cost barriers and a mental-health care cost barrier formed a particularly high-risk subgroup.

The central design principle was non-causal interpretation. This project does not claim that medication barriers, food insecurity, or mental-health care cost barriers cause depression. Instead, these variables are treated as risk markers that may help identify adults more likely to report frequent depression. This distinction matters because the appropriate policy use is supportive outreach, not diagnosis, causal labeling, or denial of services.

2. Literature Review and Research Gap

The literature review for this capstone was organized around three complementary NHIS method templates: a multi-year mental-health trend paper, an interaction/subgroup paper, and a prescription cost-barrier paper. Together, these studies show how NHIS can support trend analysis, subgroup interaction analysis, and medication-affordability measurement. The gap is that these strands are rarely combined into a prediction and calibration workflow for frequent depression.

2.1 Multi-year NHIS mental-health trends

Kim et al. (2025) used NHIS 2019-2022 data to examine trends in depression and anxiety by cancer status. The study used age-adjusted prevalence and spline logistic regression, which is useful when time trends may be nonlinear. That work demonstrated the value of pooling multiple NHIS years and evaluating mental-health outcomes over time. For the current capstone, the methodological takeaway is that pooled NHIS data can be used to study recent mental-health burden while preserving national relevance. However, the trend paper did not focus on prescription cost barriers, model trimming, prediction, or calibration.

2.2 Interaction and subgroup vulnerability

Odame et al. (2023) examined whether the association between hypertension and depression differed by citizenship status using NHIS 2015-2018 data. Their interaction approach illustrates why subgroup models can add value beyond separate descriptive tables. For the current capstone, this motivated the prescription-user extension, where the key question was whether medication barrier severity interacts with mental-health care cost barriers. However, the hypertension-citizenship study was not designed as a prediction model and did not evaluate calibration or outreach thresholds.

2.3 Prescription cost-barrier variables

Taha et al. (2022) used NHIS 2013-2018 data to study cost-related medication nonadherence among adults with diabetes. They operationalized cost-related medication behaviors such as skipping doses, taking less medication, or delaying prescription fills to save money. That study provides the closest precedent for using NHIS medication-affordability variables in a policy-relevant analysis. The current capstone reverses the direction of the outcome and predictors: rather than predicting medication nonadherence, it uses medication-access and affordability barriers as predictors of frequent depression.

2.4 Synthesis and gap

Across these three papers, several NHIS methods appear repeatedly: weighting, logistic regression, subgroup prevalence, and interaction terms. What remains missing is a workflow that connects raw NHIS feature selection, model trimming, weighted predictive performance, calibration, and public-health threshold translation. This capstone addresses that gap by building a 12-variable interpretable model, comparing it against LASSO and random forest, evaluating calibration, and translating predicted probabilities into top-risk outreach groups.

3. Methods

3.1 Data source and analytic sample

The analysis used pooled NHIS Sample Adult public-use files for 2022, 2023, and 2024. The final analytic sample included 87,809 adults with valid DEPFREQ_A responses. NHIS is a complex survey. PPSU identifies the pseudo-primary sampling unit, PSTRA identifies the pseudo-stratum, and WTFA_A is the Sample Adult final weight. Because three years were pooled, the analytic weight was WTFA_A / 3. In this project, the pooled weight was used for descriptive estimates, model fitting weights, and weighted predictive evaluation.

3.2 Outcome

The binary outcome was frequent depression. Adults were coded as 1 if DEPFREQ_A indicated depressed mood daily or weekly, and 0 if depressed mood was reported less frequently. Responses outside the valid frequency categories were excluded. The weighted prevalence of frequent depression in the pooled analytic sample was approximately 10.5%.

3.3 Candidate predictors and conceptual domains

Predictors were organized conceptually before modeling to avoid arbitrary feature fishing. The broader candidate set included 18 predictors plus timing controls. The final reporting model retained 12 human-readable predictors grouped into three domains:

Health burden: self-rated health, BMI category, smoking status, and any prescription use.

SES/access/hardship: food security, income-to-poverty group, mental-health care cost barrier, and inability to get needed prescription medication due to cost.

Demographic/social context: age group, sex, race/ethnicity, and marital status.

This domain structure also supports interpretation. Rather than treating the model as a collection of unrelated dummy variables, the final model can be described as combining health burden, material hardship, access friction, and social-demographic context.

3.4 Model development and validation

The analysis used a train/validation/test split stratified by survey year and outcome. The final presentation model used the 12 predictors only. Three 12-variable model families were evaluated: weighted logistic regression, LASSO logistic regression, and random forest. Weighted logistic regression and LASSO provide transparent linear probability-ranking models after one-hot encoding of categorical variables. Random forest was included as a flexible nonlinear comparator.

Model performance was evaluated on a held-out test set using weighted AUC, weighted Brier score, calibration slope, and expected calibration error (ECE). AUC evaluates discrimination or ranking ability: whether adults with frequent depression tend to receive higher predicted risk than adults without frequent depression. Brier score evaluates squared probability error. Calibration slope and ECE evaluate probability quality. ECE was calculated as the weighted average absolute gap between mean predicted risk and observed prevalence across risk groups.

3.5 Calibration analysis

Calibration was presented using fixed predicted-risk groups rather than weighted deciles. The risk groups were probability ranges such as 0-2.5%, 2.5-5%, 5-8%, 8-10%, 10-15%, and higher ranges. This is different from a weighted decile plot. A predicted-risk group is a fixed probability range; a weighted decile is a ranked slice of

the weighted population. The fixed risk-group approach was chosen for presentation because the x-axis directly represents predicted probability ranges, which is easier to explain to a nontechnical audience.

Two calibration-related figures were used. The first shows the percentage of the weighted adult population in each predicted-risk group. This answers where the population falls across predicted-risk groups. The second compares mean predicted risk with observed frequent-depression prevalence within each group. This answers whether predicted probabilities match observed outcomes. Calibration is considered better when those two values are close.

3.6 Policy threshold analysis

The threshold analysis translated predicted probabilities into outreach groups. Test-set adults were ranked from highest to lowest predicted risk. Survey weights were accumulated, and top-risk groups were defined as the highest-risk 10%, 20%, 30%, and so on of the weighted adult population. These percentages refer to population shares after ranking, not to risk probabilities. For example, Top 20% means the highest-risk 20% of the weighted adult population. It does not mean adults with exactly 20% risk.

For each threshold, the analysis calculated the minimum predicted-risk threshold, the observed prevalence within the selected group, and the percentage of all weighted frequent-depression cases captured. These metrics help translate a prediction model into practical outreach planning.

3.7 Prescription-user extension

The secondary extension focused on adults who used prescription medication in the past 12 months. Among prescription users, a medication barrier count was created from three binary indicators: inability to get needed prescription medication due to cost, skipped medication to save money, and delayed medication fills to save money. The count ranged from 0 to 3. The extension then compared observed frequent-depression prevalence by medication barrier count and mental-health care cost barrier status. This extension was framed as a dual-hardship risk marker rather than a causal test.

4. Results

4.1 Sample and outcome prevalence

The pooled analytic sample included 87,809 adults with valid depression-frequency data. The weighted prevalence of frequent depression was approximately 10.5%. This prevalence is high enough to justify a risk-stratification approach, but low enough that most adults should fall into lower predicted-risk groups.

4.2 Model performance

The 12-variable logistic and LASSO models performed nearly identically. LASSO logistic had weighted AUC approximately 0.787-0.788, and weighted logistic had weighted AUC approximately 0.787. Random forest was slightly lower, with weighted AUC approximately 0.783 in the final presentation workflow. Calibration metrics favored the weighted logistic model for the final presentation because it had the lowest ECE and the highest interpretability.

Table 1. Held-out performance of 12-variable models

Model	Weighted AUC	Brier	Calibration slope	ECE
LASSO logistic	0.788	0.0815	0.945	0.0115
Weighted logistic	0.787	0.0816	0.941	0.0112
Random forest	0.783	0.0828	1.261	0.0186

The results support using the 12-variable weighted logistic model for the final calibration and policy-threshold analysis. The tiny AUC advantage of LASSO was not large enough to outweigh the interpretability of standard weighted logistic regression. Random forest did not provide a meaningful discrimination gain and had higher calibration error.

4.3 Calibration by fixed predicted-risk groups

The predicted-risk distribution showed that most adults were assigned to lower risk groups. This is expected given the overall weighted prevalence of approximately 10.5%. The calibration comparison showed that predicted risk and observed prevalence were close in many low-to-middle risk groups, while the largest discrepancies appeared in the highest-risk ranges. This indicates that the model is useful for ranking and risk stratification, but probability thresholds should be interpreted as approximate and should be calibrated or externally validated before operational use.

Figure 1. Weighted population percentage across predicted-risk groups.

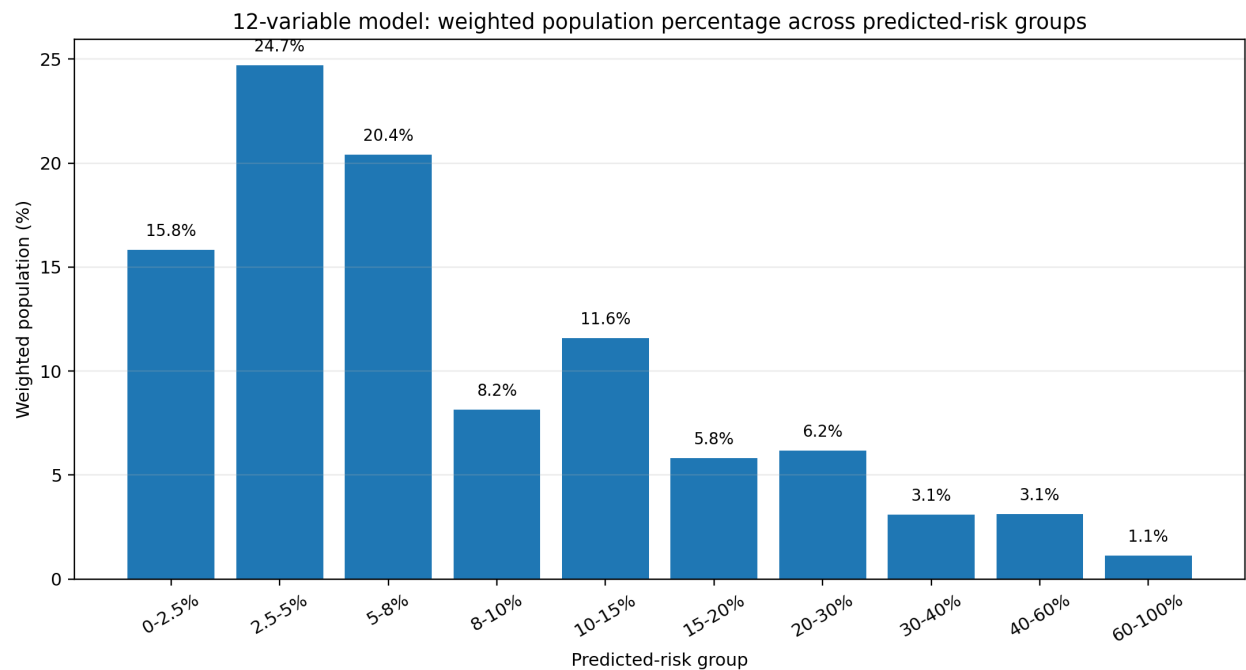
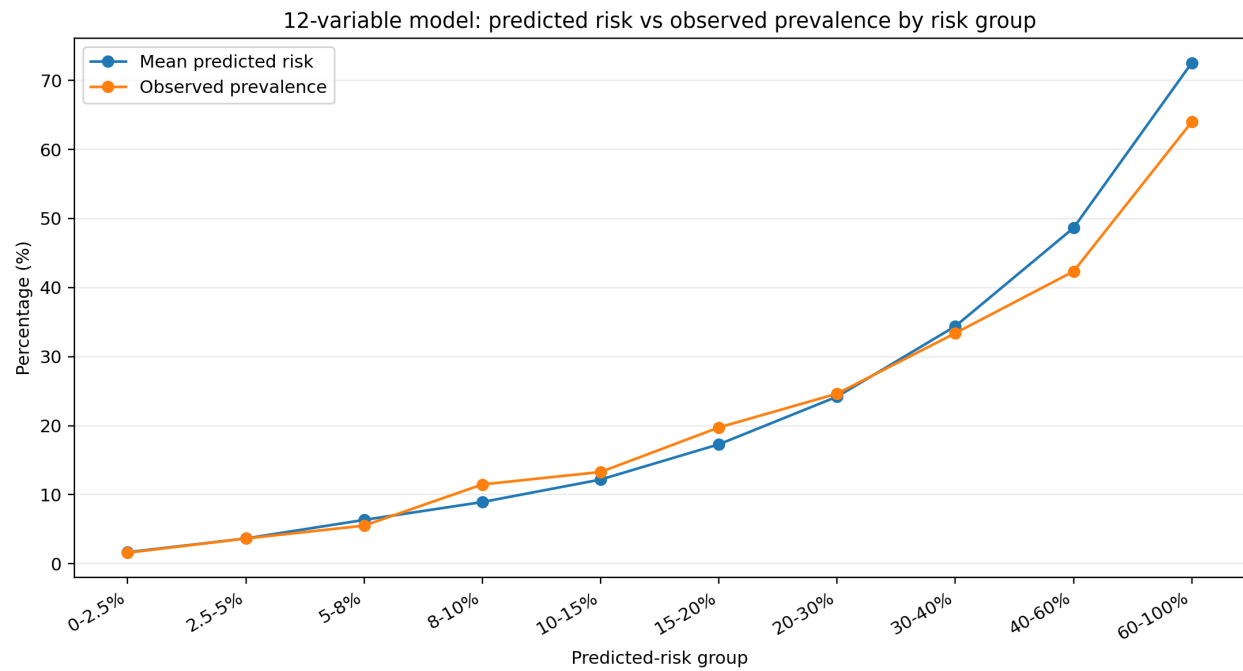


Figure 2. Mean predicted risk versus observed prevalence by predicted-risk group.



4.4 Outreach thresholds and case capture

The threshold analysis shows how predicted probabilities can be translated into outreach intensity. The highest-risk 10% of the weighted adult population had a predicted-risk threshold of approximately 24.4%, captured approximately 35.8% of weighted frequent-depression cases, and had observed prevalence approximately 37.6%. Expanding to the top 20% lowered the threshold to approximately 14.7%, captured approximately 56.0% of cases, and had observed prevalence approximately 29.4%. Expanding to the top 30% lowered the threshold to approximately 10.3%, captured approximately 68.2% of cases, and had observed prevalence approximately 23.9%.

Table 2. Outreach thresholds from the 12-variable weighted logistic model

Selected group	Risk threshold	Case capture	Observed prevalence
Top 10%	24.4%	35.8%	37.6%
Top 20%	14.7%	56.0%	29.4%
Top 30%	10.3%	68.2%	23.9%

As the selected top-risk group expands from 10% to 100% of the weighted population, observed prevalence declines because lower-risk adults are progressively added. At the same time, case capture increases because more total cases are included. These two curves clarify the tradeoff between intensive targeting and broader coverage.

Figure 3. Observed prevalence declines as the selected top-risk group expands.

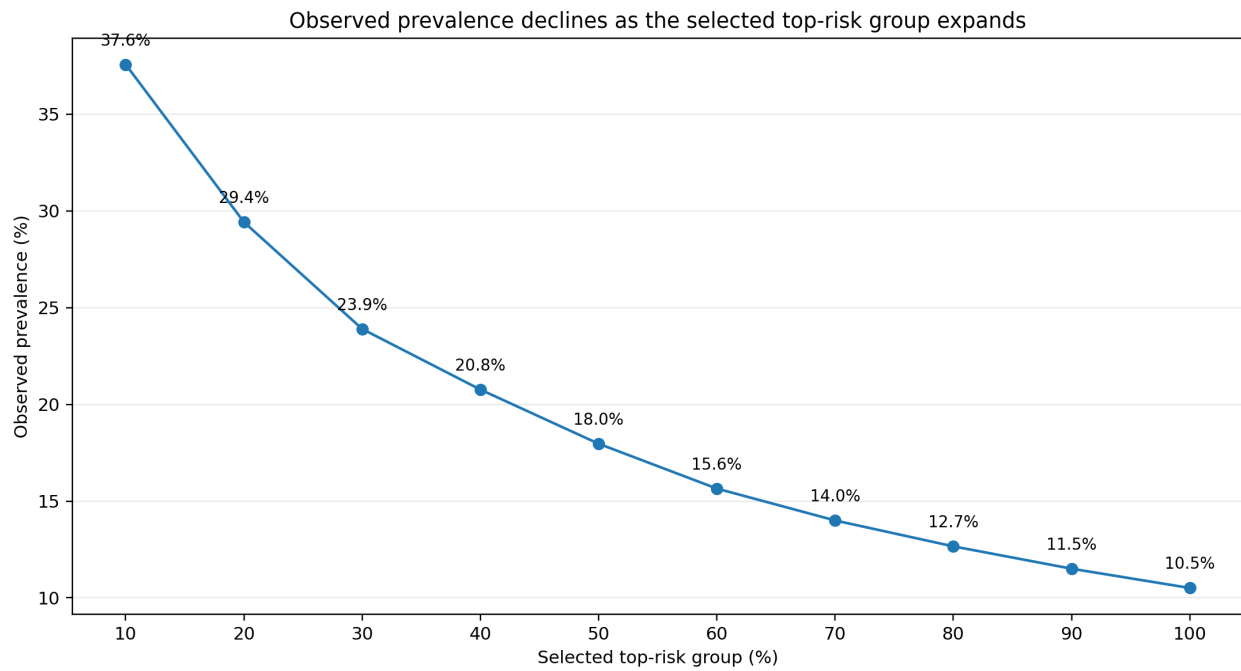
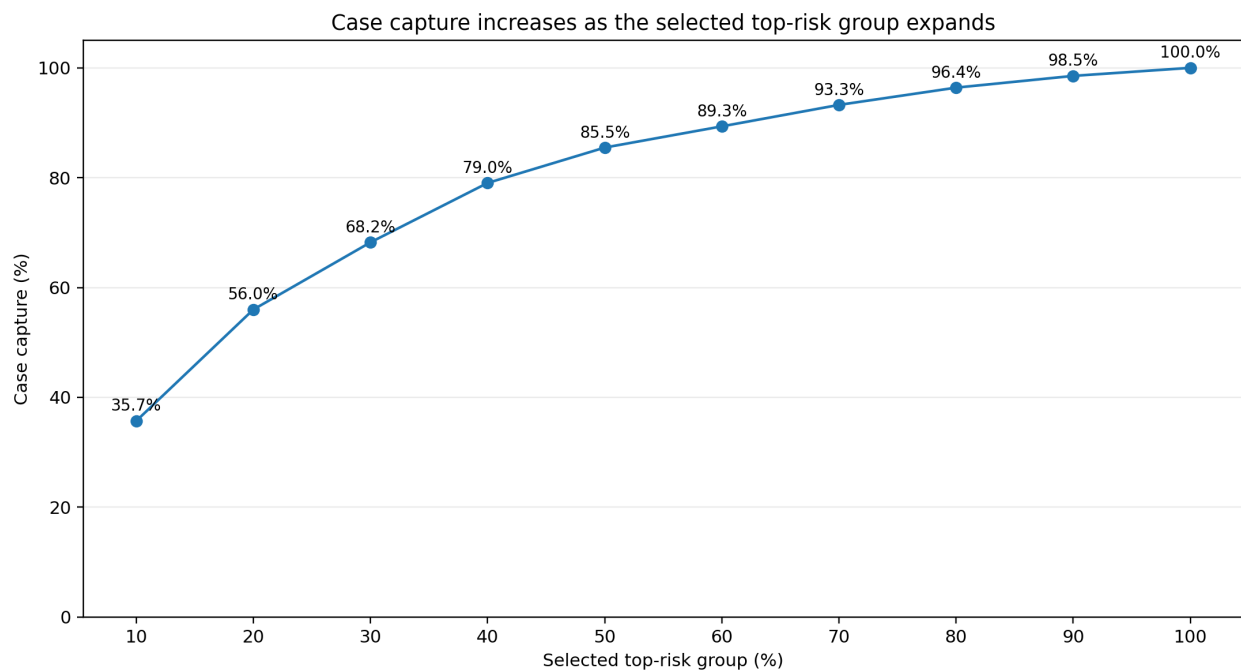


Figure 4. Case capture increases as the selected top-risk group expands.



4.5 Prescription-user dual-barrier extension

Among prescription users, frequent-depression prevalence increased with medication barrier count. The pattern was stronger among adults who also reported a mental-health care cost barrier. For prescription users without a mental-health care cost barrier, observed prevalence increased from approximately 10.1% at zero medication

barriers to approximately 24.9% at three medication barriers. For prescription users with a mental-health care cost barrier, observed prevalence increased from approximately 31.8% at zero medication barriers to approximately 57.9% at three medication barriers.

Table 3. Rx-user observed prevalence by medication barrier count and mental-health care cost barrier

Mental-health care cost barrier	0 barriers	1 barrier	2 barriers	3 barriers
No MH care cost barrier	10.1%	20.4%	22.2%	24.9%
MH care cost barrier	31.8%	41.9%	44.4%	57.9%

This pattern supports a policy-relevant dual-hardship interpretation. The extension is not evidence that medication barriers cause depression; rather, it shows that overlapping medication and mental-health affordability barriers identify a subgroup with high observed frequent-depression prevalence.

5. Discussion

This capstone developed a coherent, interpretable, and calibration-aware prediction workflow for frequent depression using pooled NHIS 2022-2024 Sample Adult data. The final model was deliberately simple: a 12-variable weighted logistic model. The model performed nearly as well as LASSO and slightly better than random forest in the final trimmed feature set. This supports the conclusion that the 12 human-readable predictors already capture much of the available signal in the public-use NHIS variables.

The main improvement over an AUC-only workflow is the emphasis on calibration. AUC answers whether the model ranks higher-risk adults above lower-risk adults. Calibration answers whether the predicted probabilities match observed prevalence. This distinction is essential for public-health translation. If a model is only used for broad ranking, AUC may be sufficient. If a model is used to define outreach thresholds, calibration becomes more important because programs need to know whether a predicted risk of 20% corresponds approximately to 20% observed prevalence.

The threshold analysis provides a practical translation. The top 10% group is concentrated and has high observed prevalence, so it could support more intensive outreach or navigation. The top 20% or top 30% groups capture more cases and could support broader screening invitations or care-navigation efforts. The declining prevalence and increasing case-capture curves show the tradeoff between precision and coverage. This framing is more realistic than presenting a single threshold as universally optimal.

The prescription-user extension adds a secondary public-health story. Medication barrier severity alone is a strong risk marker, but the highest observed prevalence appears when medication barriers overlap with mental-health care cost barriers. This suggests that programs addressing medication affordability and mental-health access should not be siloed. Adults facing both barriers may benefit from coordinated care navigation, benefits counseling, and screening support.

6. Policy Implications

The final model is best understood as a supportive prioritization tool. Potential uses include targeted screening invitations, proactive outreach, navigator referral, benefits counseling, and population-level resource planning. The model should not be used to diagnose depression, assign causal blame, deny care, or determine eligibility for services. High predicted risk should trigger support, not punishment.

For a public-health department or health system, the model could be used to define tiers of outreach. The highest-risk 10% could receive more intensive follow-up, while the top 20% or top 30% could receive screening reminders, mental-health resource information, or referral options. The Rx-user dual-barrier subgroup could be flagged for medication affordability support and mental-health care navigation.

7. Limitations

Several limitations should be emphasized. First, NHIS is cross-sectional, so the analysis supports prediction and association, not causality. Second, the outcome is based on self-reported depressed mood frequency rather than a clinical diagnostic interview. Third, predictors such as cost barriers, food security, and self-rated health are self-reported and may be affected by recall or reporting differences. Fourth, the public-use NHIS variables contain only the information available in the Sample Adult files, so unmeasured clinical, neighborhood, and longitudinal factors are not included. Fifth, machine-learning evaluation with complex survey data remains approximate when implemented through standard predictive modeling packages. Finally, the model was internally evaluated using a held-out split, but external validation in future NHIS years or other datasets would be needed before operational deployment.

8. Conclusion

This project produced an interpretable, survey-weighted risk-stratification workflow for frequent depression among U.S. adults. A 12-variable weighted logistic model achieved strong discrimination, reasonable calibration, and a policy-relevant threshold structure while remaining easy to explain. The strongest predictive signals reflect health burden, socioeconomic hardship, access barriers, and demographic/social context. The prescription-user extension identified a dual-barrier subgroup with especially high observed frequent-depression prevalence. The final contribution is not a causal explanation of depression, but a reproducible and calibration-aware framework for using NHIS variables to support outreach, screening, navigation, and resource planning.

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Appendix: Reproducibility Notes

The accompanying GitHub-ready script is `nhis_capstone_pipeline.py`. It expects `adult22.csv`, `adult23.csv`, and `adult24.csv` in the repository root.

The script reads only the variables required for this capstone, applies pooled weight $WTFA_A / 3$, reproduces the 12-variable model comparison, exports model metrics, and saves calibration and threshold figures.

Raw NHIS CSV files should not be committed to a public GitHub repository. The included `.gitignore` excludes data files and generated outputs by default.